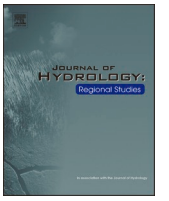
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Observing the recent floods and drought in the Lake Victoria Basin using Earth observations and hydrological anomalies

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ABSTRACT

The Lake Victoria Basin, home of Africa's largest freshwater lake, experienced extensive floods in 2019–2020 and anomalous drought conditions in early 2022. Both antithetical hydrological events raised concerns for an estimated 35 million people within the basin who are vulnerable to these continually recurrent hydrometeorological extremes. While there is a need for high-resolution spatial and temporal hydrological monitoring, in-situ observations are limited and insufficient within the basin. This study assesses the capability of publicly available Earth observations and models to capture the occurrence of recent hydrological extremes within the Lake Victoria Basin. Spatial and temporal comparisons of the following hydrological variables are conducted within the Lake Victoria Basin from June 2002 – June 2022: precipitation, evapotranspiration, runoff, soil moisture, lake height, and total water storage anomaly (TWSA). Results from the time series and spatial analyses confirm that the anomalous wet and dry events respectively occurred from October 2019 – March 2020 and November 2021 – March 2021. The observations also indicated a quick recovery from the drought event by June 2022. Additionally, a comparison of 9-km soil moisture and 1-km downscaled near-surface soil moisture products improves understanding of wetness spatial variability within the basin. These results were validated using in-situ soil moisture data available through the International Soil Moisture Network, making this the first study to validate the downscaled 1-km soil moil moisture product within the African continent. Methods from this study demonstrate the utility of a diverse array of publicly available Earth observation and model data to improve understanding of recent hydrological anomalies and extreme events within the Lake Victoria Basin. This can be applied to water resources management, particularly in regions lacking sufficient in-situ hydrological data.

1. Introduction

Africa's largest lake, Lake Victoria, provides valuable ecosystem services to five East African countries within its basin: Burundi, Kenya, Rwanda, Tanzania, and Uganda. Freshwater from Lake Victoria supports domestic use, diverse aquatic and terrestrial ecosystems, and economic industries including agriculture, livestock, fisheries, and hydropower production (Awange, 2020, 2006). Tourism, recreation, and transportation of goods via inland waterways are further supported by the lake (Awange, 2020; Reynolds, 2005). As the source of the White Nile River and a subbasin of the greater Nile watershed, the Lake Victoria Basin plays a critical role in supplying freshwater locally to an estimated 35 million people and indirectly to 340 million people in downstream countries

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(Ramillien et al., 2014; Mugo et al., 2020).

Over the previous decades, hydrometeorological extremes such as floods and droughts have proven to be continual and recurrent in the Lake Victoria Basin (Nyeko-Ogiramoi et al., 2013). Historically, devastating floods in this region occur practically every year during the October-December rainy season and cause substantial loss of life and property (Nyeko-Ogiramoi et al., 2013). Periods of extreme drought are equally concerning because decreased or delayed rainfall can lead to crop failure, widespread famine, and food insecurity (Coughlan de Perez et al. (2019); Agutu et al. (2017); FEWS NET (2021)). As the frequency and severity of hydrological extremes is predicted to increase due to climate change, accurate and continual monitoring of hydrological conditions at the basin level is vital for sustainable water resources management (Alsdorf et al., 2016; Paris et al., 2022; Wainwright et al., 2021). This is particularly true for the Lake Victoria Basin, particularly given that previous research has identified it as a significant hotspot for population-climate burdens which are exacerbated by recurrent droughts and flooding due to variable rainfall (López-Carr et al., 2014). Since the populations of Burundi, Kenya, Rwanda, Tanzania, and Uganda are respectively forecasted to increase by 114%, 68%, 77%, 117%, and 114% between mid-2020 and mid-2050, there is further motivation to improve water resources management to support such population growth and demands (Table 1) (Kaneda et al., 2020).

As with many regions in Africa, the Lake Victoria Basin lacks sufficient in-situ, public, and ground-based hydrological observations to meet these needs (Khan et al., 2010; Hu et al., 2021; Ndehedehe et al., 2016). While discharge data from Uganda's hydropower dam located at Lake Victoria's outlet is collected, this data is not publicly available. This is likely due to political concerns regarding the Agreed Curve, which is an international treaty between Uganda and Egypt through which Uganda agrees to operate its hydropower dams in a way that reflect the natural outflows of Lake Victoria (Swenson and Wahr, 2009; Shamsudduha et al., 2017; Kull, 2006; Vanderkelen et al., 2018; Getirana et al., 2020). Similarly, the few ground-based rainfall stations within the Lake Victoria Basin are unreliably operated and sparsely distributed, making it an insufficient data source for basin-wide hydrological analyses (Awange et al., 2019). Therefore, there is a need for publicly available and remotely observed hydrological data at sufficient spatial and temporal resolutions in the Lake Victoria Basin to monitor the recent hydrological extreme events.

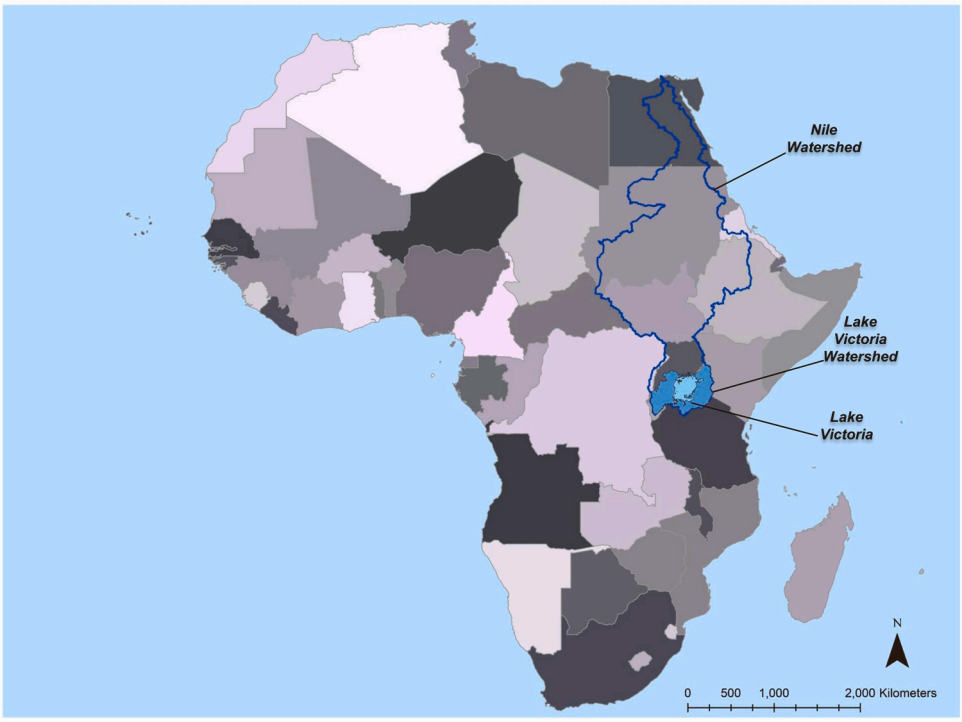
Satellite remote sensing has proven to be a useful tool to obtain hydrological observations in regions limited by sparse temporal and spatial coverage of ground-based in-situ data. This is particularly true for comparisons of satellite sensor precipitation with gauge observations (Hashemi et al., 2017; Mondal et al., 2018; Le et al., 2018). Numerous studies have relied on remote sensing and land surface models to reliably estimate components of the hydrological cycle within the Lake Victoria Basin and other African basins (Lakshmi et al., 2018; Khan et al., 2010; Awange, 2020; Ndehedehe et al., 2016; Vanderkelen et al., 2018; Becker et al., 2010; Grippa et al., 2011; Shamsudduha et al., 2017; Kansara et al., 2021). Other studies in the Lower Mekong River Basin have also demonstrated the synergistic utility of satellite remote sensing observations within hydrological models (Mohammed et al., 2018a, 2018b, 2018c; Le et al., 2020, 2022). These techniques, particularly the utilization of a wide array of publicly available hydrological Earth observations, can be applied to the Lake Victoria Basin to update previous studies and to compare the recent hydrological extreme events. Additionally, this is a valuable method for collecting hydrological data across country borders since Lake Victoria is geopolitically divided between Kenya (6%), Uganda (43%), and Tanzania (51%) (Awange, 2020). Thus, Earth observations are a promising method for obtaining uniform and publicly available hydrological observations at sufficient spatial and temporal resolutions across the international boundaries of this basin.

In this paper, we examine components of the hydrological water balance (precipitation, evapotranspiration, runoff, and change in total water storage) and additional hydrological variables (soil moisture, lake level height, and lake extent) for the Lake Victoria Basin from June 2002 – June 2022 with particular focus on the 2019–2020 flood events and the 2021–2022 drought period. This is accomplished by leveraging publicly available data from seven satellite-based Earth observations and one land surface model, which is necessary given the limited in-situ data available in this region. Since previous studies identified the Lake Victoria Basin as an area where populations are particularly vulnerable to seasonal hydrological variabilities, we conducted a hydrological anomaly analysis at monthly timescales to identify periods of anomalous positive and negative hydrological conditions. Our analysis includes monthly basin-wide averaged time series of precipitation, evapotranspiration, runoff, total water storage anomaly, soil moisture, and lake level height from June 2002 – June 2022. Spatial anomaly plots at spatial resolution up to 0.1° are provided for precipitation, evapotranspiration, runoff, total water storage anomaly, and soil moisture for two months of interest: the local maxima and minima monthly anomalies between January 2019 – June 2022. These represent a wet and dry month which respectively occurred during the identified flood and drought events. To our knowledge, this is the first study to conduct a hydrological spatial anomaly comparison for the 2019–2020 flood and 2021–2022 drought events in the Lake Victoria Basin.

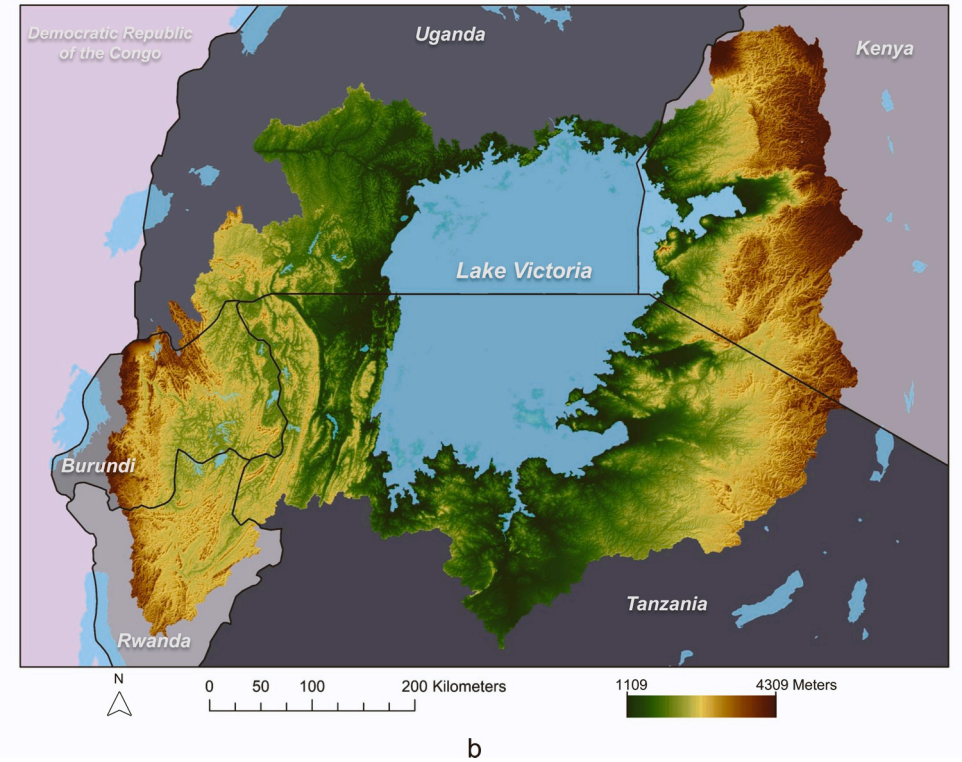
Additionally, to improve understanding of wetness spatial variability during the flood and drought periods, a spatial comparison of 9-km near-surface soil moisture and 1-km downscaled near-surface soil moisture products is included for the basin. These spatial plots

Table 1
Population forecasts for Burundi, Kenya, Rwanda, Tanzania, and Uganda from mid-2020 to mid-2050 (Kaneda et al., 2020).

Country	Mid-2020 population (millions)	Mid-2035 population (millions)	Mid-2050 population (millions)	Percent population change from 2020 to 2050 (%)
Burundi	11.9	18.3	25.5	+114%
Kenya	53.5	71.9	89.7	+68%
Rwanda	13.0	17.9	23.0	+77%
Tanzania	59.7	90.4	129.4	+117%
Uganda	45.7	69.5	97.7	+114%



a



b

(caption on next page)

Fig. 1. (a) Map of watershed delineations for the Nile and Lake Victoria watersheds. Lake Victoria is the source of the White Nile River and thus plays a critical role in the hydrology of downstream countries within the Nile watershed. Country boundaries are provided to give context to the watershed at the continental scale (ICPAC, 2019; World Bank, 2019; Esri, 2021; WWF, 2019). (b). Digital elevation model (DEM) for the Lake Victoria Basin, derived from the Shuttle Radar Topography Mission (SRTM). Delineation of the basin includes portions of Uganda, Kenya, Tanzania, Rwanda, and Burundi. Major lakes are displayed in light blue. The highest topography is found in the northwestern portion of the basin in Kenya and along the southwestern rim in Burundi and Rwanda. The lowest topography is found along the perimeter of Lake Victoria (USGS, 2018; ICPAC, 2019; World Bank, 2019; Esri, 2021; WWF, 2019).

are displayed for the wet and dry months identified in the root zone soil moisture anomaly time series as local maxima and minima between January 2019 – June 2022. The 1-km downscaled near-surface soil moisture data was validated using 28 ground soil moisture stations available via the International Soil Moisture Network. This is the first study to validate the 1-km downscaled near-surface soil moisture product within the African continent. Because the Lake Victoria Basin and other areas in the Greater Horn of Africa lack sufficient in-situ soil moisture observations for widespread analysis, validation and assessment of remotely sensed and modeled soil moisture data are vital for determining the utility of these data for future studies and operational usage (Agutu et al., 2021). This is especially necessary for soil moisture due to the region's dependence on rain-fed subsistence agriculture (Agutu et al., 2021).

Methods from this study for observing hydrological anomalies as time series and spatial plots demonstrate a simple tool for monitoring hydrological variables using publicly available datasets in a region that is traditionally limited by insufficient ground observations. These methods could be applied to inform water management and agricultural planning within the Lake Victoria Basin, as well as other regions lacking sufficient ground-based hydrological observations. Further research could be conducted to assess applications of this work for water management. Here, we show that Earth observations can monitor hydrological variables during extreme and anomalous events, such as the 2019–2020 flood and 2021–2022 drought. This work aims to support ongoing efforts to decrease the community's vulnerability to anomalous hydrological conditions, such as extreme floods and droughts, by improving monitoring of components of the hydrological cycle (precipitation, evapotranspiration, runoff, and change in total water storage) and other hydrological variables (soil moisture, lake level height, and lake extent) within the Lake Victoria Basin.

2. Study area and data

2.1. Study area

Lake Victoria, locally called *Lake Nulabale,* is the largest tropical lake and the second largest freshwater lake in the world with an approximate surface area of 65,295 km^2 (Awange, 2020). Despite its large spatial extent, Lake Victoria is relatively shallow with a maximum depth of 80 m and an average depth of 40 m (Nyeko-Ogiramoi et al., 2013). The basin has a catchment surface area of approximately 229,815 km^2 and encompasses portions of Uganda, Kenya, Tanzania, Rwanda, and Burundi (Hamilton, 2016). A singular outlet of Lake Victoria is found at Ripon Falls near Jinja, Uganda. Here, the Lake Victoria Basin directly contributes to the White Nile River. Thus, as shown in Fig. 1A, the Lake Victoria Basin is a subbasin of the Nile Watershed.

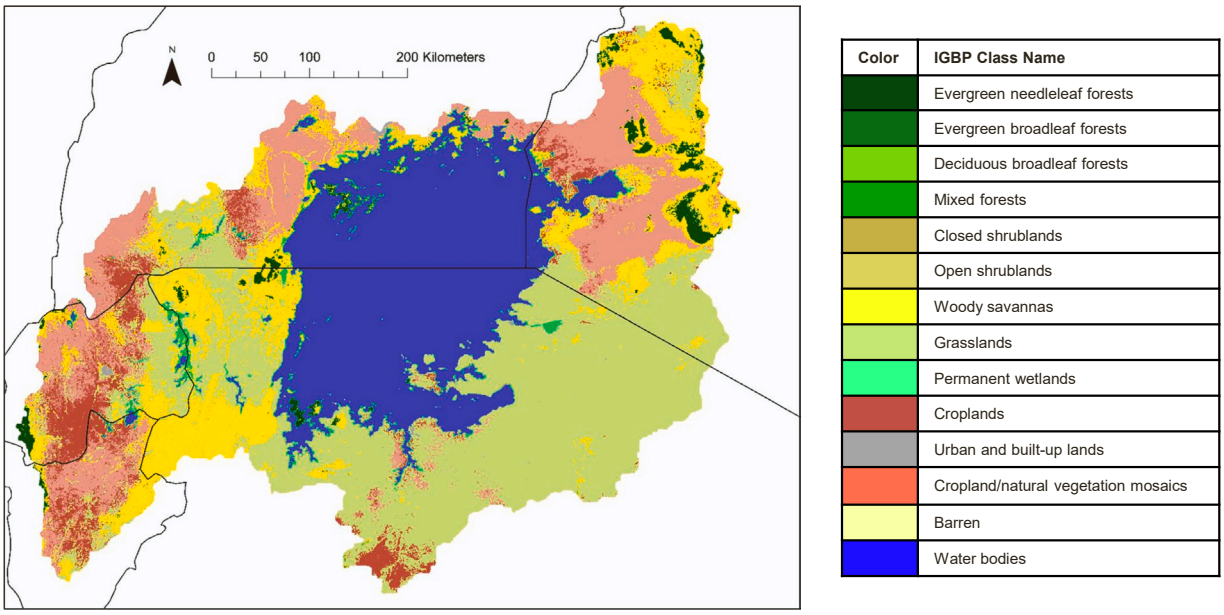


Fig. 2. Land cover classifications for the Lake Victoria Basin in 2019 derived from MODIS/Terra + Aqua Yearly L3 Global 500 m SIN GRID (MCD12Q1 V006). The classifications are based on the Annual International Geosphere-Biosphere Program (IGBP) classification scheme.

A digital elevation model (DEM) of the Lake Victoria Basin derived from the Shuttle Radar Topography Mission (SRTM) is shown in Fig. 1B (USGS, 2018). Tectonic and volcanic activity within the Great Rift Valley contribute to the basin's varying topography which ranges in altitude from approximately 1109 – 4309 m above mean sea level (MSL). The highest elevations are found in the northeastern mountainous regions in Kenya and along the southwestern edge of the basin in Burundi and Rwanda. The lowest elevations are found directly along the perimeter of Lake Victoria.

Approximately 25 major rivers within the basin contribute directly to Lake Victoria (Vanderkelen et al., 2018). However, precipitation accounts for approximately 86% of the input into Lake Victoria (Nyeko-Ogiramoi et al., 2013). The humid, tropical climate in combination with Lake Victoria's large extent contribute to a distinct diurnal meteorological pattern within the basin. During afternoons, intense convective thunderstorms typically form over the land due to a nocturnal lake-land breeze system (Shamsudduha et al., 2017; Thiery et al., 2016, 2017). On seasonal timescales, the *long rains* and *short rains* respectively occur from May – March and September – November. Hydrological patterns within the basin are further influenced on seasonal and interannual scales by the Intertropical Convergence Zone (ITCZ), the Indian Ocean Dipole (IOD), and the El-Ñino Southern Oscillation (ENSO) (Wainwright et al., 2021; Becker et al., 2010; FEWS NET, 2020, 2021).

2.1.1. Land cover of the Lake Victoria Basin

Land cover classifications derived from MODIS Terra and Aqua reflectance data at a spatial resolution of 500 m are shown for the Lake Victoria Basin in Fig. 2. For purposes of this study, land cover is assumed to be constant between June 2019 to June 2022 since analysis of land cover change were outside the scope of this study.

Inland waterbodies, including Lake Victoria, make up 25% of the basin's total area. Grasslands are 35% of the basin area and are predominately located in Tanzania. Croplands (including natural vegetation mosaics) are collectively 21% of the basin extent and include portions of all five countries. The main crops produced within the basin are maize, beans, sorghum, millet, paddy rice, sugarcane, tea, coffee, cotton, and meats (Reynolds, 2005; Mugo et al., 2020).

Only 0.2% of the basin area is classified as urban and built-up lands even though an estimated 35 million people live within a 100 km buffer zone of Lake Victoria (Mugo et al., 2020). A study conducted by Mugo et al. (2020) used optical imagery from Landsat and field data to assess land use changes between 1985 and 2014 in the Lake Victoria Basin and found an 800% increase in urbanized areas between 1985 and 2020 (Mugo et al., 2020). Human-centric activities were found to be the primary motivation for land use land cover changes during this period, particularly through the conversion of forests, woodlands, grasslands, and wetlands to farmlands and settlements (Mugo et al., 2020).

2.2. Data

A summary table of the satellite-based Earth observations and model outputs used in this study for precipitation,

Table 2
Summary of hydrological data used in this study.

Hydrological Variable	Sensor/Model	Spatial Resolution	Temporal Resolution	Study period	Sources
Precipitation	GPM IMERG Final Precipitation L3 V06	0.1°	1 month	Jun. 2002 – Oct. 2021	Huffman et al. (2015)
	GPM IMERG Late Precipitation L3 V06	0.1°	1 day	Nov. 2021 – Jun. 2022	Huffman et al. (2015)
Evapotranspiration	MODIS MOD16A2.006	0.05°	8 days	Jun. 2002 – Jan. 2022	Running et al. (2017)
Land Cover	MODIS MDC12Q1.006	0.05°	1 year	2019	Friedl et al. (2019)
Runoff	GLDAS Noah Land Surface Model L4 V2.1	0.25°	1 month	Jun. 2002 – Apr. 2022	Rodell et al. (2004)
	GLDAS Noah Land Surface Model L4 Early Product V2.1	0.25°	1 month	May. 2022 – Jun. 2022	Rodell et al. (2004)
Root Zone Soil Moisture (0 – 2 m depth)	GLDAS Noah Land Surface Model L4 V2.1	0.25°	1 month	Jun. 2002 – Apr. 2022	Rodell et al. (2004)
	GLDAS Noah Land Surface Model L4 Early Product V2.1	0.25°	1 month	May. 2022 – Jun. 2022	Rodell et al. (2004)
Near-surface Soil Moisture (0–5 cm depth)	SMAP	0.09°	1 day	Jan. 2015 – Jan. 2022	Entekhabi et al. (2010)
	SMAP	0.01°	1 day	Jan. 2015 – Jan. 2022	Fang et al. (2022)
Lake Height	Jason-1/USDA G-REALM	$> 100\ km^2$	10 days	Jun. 2002 – 2008	USDA G-REALM (2022)
	Jason-2/USDA G-REALM	$> 100\ km^2$	10 days	2008 – 2016	USDA G-REALM (2022)
	Jason-3/USDA G-REALM	$> 100\ km^2$	10 days	2016 – Jun. 2022	USDA G-REALM (2022)
Total Water Storage Anomaly	GRACE/GRACE-FO Tellus Monthly Mass Grid Global Mascons (JPL RL06 V02)	0.5°	1 month	Jun. 2002 – May. 2022	Landerer and Swenson (2012)

evapotranspiration, runoff, soil moisture (root zone and near-surface), lake height, and total water storage anomaly (TWSA) are summarized in Table 2. Each variable was averaged at a monthly timescale in the native resolution of the source data. Monthly spatial averages were calculated across the basin extent to construct time series for each variable. For spatial mapping purposes, precipitation, evapotranspiration, runoff, and root zone soil moisture were re-gridded to a common spatial resolution. The near-surface soil moisture observations were not gridded to allow for spatial comparison of the 9-km and 1-km products. Similarly, TWSA was not re-gridded to avoid misleading visual representation of these data given the coarse spatial resolution of GRACE. Lake height is assumed to be constant across the extent of Lake Victoria and thus is not spatially mapped. Further details concerning each data product can be found in Sections 2.2.1–2.2.6.

2.2.1. Precipitation data from the Integrated Multisatellite Retrievals for Global Precipitation Measurement (GPM IMERG)

The Integrated Multisatellite Retrievals for Global Precipitation Measurement (GPM IMERG) is a joint mission between the National Aeronautics Space Administration (NASA) and the Japan Aerospace and Exploration Agency (JAXA) which uses a constellation of satellites with passive microwave and infrared sensors to provide high spatial and temporal resolution precipitation measurements (Huffman et al., 2015). For regions lacking ground-based in-situ precipitation measurements, remotely sensed precipitation observations are commonly used from GPM IMERG, the Tropical Rainfall Measurement Mission (TRMM), and the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) (Hou et al., 2014; Huffman et al., 2015; Dezfuli et al., 2017). For purposes of this study, GPM IMERG Final Precipitation L3 V06 and GPM IMERG Late Precipitation L3 V06 were used to obtain monthly precipitation data at 0.1° spatial resolution from June 2002 – June 2022. This was used to construct time series and spatial plots of monthly precipitation anomalies. More information about the GPM IMERG data set can be found at https://gpm.nasa.gov/documents/IMERG-V06-Technical-Documentation.

2.2.2. Evapotranspiration and land cover data from the Moderate Resolution Imaging Spectroradiometer (MODIS)

The Moderate Resolution Imaging Spectroradiometer (MODIS) is a sensor onboard the sun-synchronous polar orbiting satellites, Terra and Aqua. In this study, MODIS was used to obtain monthly evapotranspiration data from June 2002 to June 2022 and one year of land cover data (2019). The global MODIS evapotranspiration product is based on the Penman-Monteith equation (Mu et al., 2007, 2011). This product has a temporal repeat of 8 days and a spatial resolution of 0.05° (Running et al., 2017). For this study, the MODIS MOD16A2.006 evapotranspiration data were averaged at monthly timescales to construct a time series and spatial plots of monthly evapotranspiration anomalies. It is important to note that the MODIS MOD16A2.006 evapotranspiration product masks inland water. Thus, a limitation of this observation data source is that no evapotranspiration data is available directly over Lake Victoria. The MODIS MCD12Q1.006 product was used to construct a map of land cover type for the Lake Victoria Basin in 2019 using the annual International Geosphere-Biosphere Program (IGBP) classification scheme (Friedl et al., 2019). Additional information about MODIS evapotranspiration and land cover products can be found at https://modis.gsfc.nasa.gov.

2.2.3. Runoff and root zone soil moisture data from the Global Land Data Assimilation System (GLDAS)

The Global Land Data Assimilation System (GLDAS) is a global, high resolution (0.25°), publicly available hydrological model that assimilates ground and satellite-based observations in offline land surface models to output land surface states and fluxes (Rodell et al., 2004). For hydrological basins lacking continual and extensive in-situ observations, GLDAS model outputs have performed well when estimating hydrological variables for water balance studies (Lakshmi et al., 2018; Shamsudduha et al., 2017; Kansara et al., 2021; Hu et al., 2021; Ndehedehe et al., 2016). For purposes of this study, the GLDAS Noah land surface model was used to obtain monthly runoff and root zone soil moisture (0 – 2 m depth) data. These data were used to construct time series and spatial plots of monthly runoff and root zone soil moisture anomalies. The GLDAS Noah Land Surface Model L4 V2.1 was used from June 2002 – April 2022 and the GLDAS Noah Land Surface Model L4 Early Product V2.1 was used from May 2022 – June 2022. Further details about GLDAS and the Noah land surface model can be found at https://ldas.gsfc.nasa.gov/gldas.

2.2.4. Soil moisture data from the Soil Moisture Active Passive (SMAP) mission

The Soil Moisture Active Passive (SMAP) mission launched in 2015 to estimate near-surface soil moisture (0–5 cm depth) using a 1.41 GHz L-band radiometer (Chan et al., 2016; Entekhabi et al., 2010). SMAP overpasses occur at 6:00AM and 6:00PM local time every 2–3 days with a spatial resolution on 0.09°. For this study, monthly 9-km SMAP soil moisture data from the 6:00PM overpasses were used from January 2015 – January 2022. A 1-km soil moisture product, which was downscaled from the 9-km SMAP data using correlations between surface temperature differences and vegetation, were averaged to a monthly timescale (Fang et al., 2022). Validation of the 1-km soil moisture product were performed using data available from 28 ground stations via the International Soil Moisture Network (https://ismn.geo.tuwien.ac.at). Additional information about SMAP and the 1-km downscaled soil moisture product can be found at https://smap.jpl.nasa.gov and Fang et al. (2022). Due to the limited observation record of SMAP of 8 years, this study uses observed monthly values of 9-km and 1-km near-surface soil moisture data as opposed to monthly anomalies. These data were used to construct time series and spatial plots of monthly near-surface soil moisture at 9-km and 1-km resolutions.

2.2.5. Lake level height data from satellite radar altimetry (Jason-1, Jason-2, Jason-3)

Satellite radar altimetry has proven to be a reliable method for monitoring inland waterbody elevation over large lakes, rivers, and wetlands, particularly in regions lacking adequate stream gage data (Ramillien et al., 2014; Becker et al., 2010; Ndehedehe et al., 2016; Birkett et al., 2010). To calculate lake surface height, a nadir-pointing antenna onboard the satellite emits microwave pulses and measures the elapsed time between emission and receiving of the reflected pulse. Altitude, elapse time, and geophysical corrections are

used to estimate lake surface height. The microwave spectra enable measurements regardless of day/night or cloudy conditions.

Since stream gage data at Lake Victoria's outlet is not publicly available, satellite radar altimetry data obtained from the United States Department of Agriculture (USDA) Foreign Agricultural Service Global Reservoir and Lake Monitor (G-REALM) were used to estimate lake level height in millimeters relative to MSL (USDA, 2022). Previous studies have also utilized G-REALM altimetry data for analysis within the Lake Victoria Basin (Swenson and Wahr, 2009; Becker et al., 2010; Reynolds, 2005). A validation study was conducted by Reynolds in 2005 that compared Lake Victoria gage data from 2000 to 2004 to G-REALM and found very similar agreement (better than 10 cm RSM) (Reynolds, 2005). In our study, the following satellites and sensors with an approximate repeat period of 10 days were used from the G-REALM Lake Victoria height product to construct a time series of monthly lake height anomalies: Jason-1 GDR 20 Hz from 2002 to 2008, Jason-2/OSTM GDR from 2008 to 2016, and Jason-3 Interim GDR 20 Hz from 2016 to 2022. Further details about G-REALM can be found at https://ipad.fas.usda.gov/cropexplorer/global_reservoir/.

2.2.6. *Total water storage anomaly data from the Gravity Recovery and Climate Experiment (GRACE/GRACE-FO)*

Satellite gravimetry via the Gravity Recovery and Climate Experiment (GRACE) and Follow-On (GRACE-FO) missions has proven to be a reliable method for observing changes in total water storage with a geoid height accuracy of one centimeter (Ramillien et al., 2020). Changes in catchment storage from aquifers, soil moisture, groundwater, surface water, etc. are observable as GRACE-derived total water storage anomaly (TWSA). Previous studies have utilized GRACE TWSA for hydrological balance analysis in the Lake Victoria Basin and surrounding African basins (Awange et al., 2008; Lakshmi et al., 2018; Ndehedehe et al., 2016; Khaki and Awange, 2021; Kansara et al., 2021; Grippa et al., 2011; Agutu et al., 2017). In this study, the GRC Tellus JPL Monthly Mass Grid Global Mascons (JPL RL06 V02) data with 0.5° spatial resolution were used (please see the GRACE-FO L3 Handbook: https://podaac-tools.jpl.nasa.gov/drive/files/allData/gracefo/docs/GRACE-FO_L3_Handbook_JPL.pdf). While the GRC Tellus JPL Monthly Mass Grid Global Mascon data is provided at this spatial resolution of 0.5°, it is important to note that the size of a single mascon is 3° and neighboring 0.5° grid cells may not be independent of each other. However, it is common to use the GRC Tellus JPL Monthly Mass Grid Global Mascon dataset at 0.5° resolution (Kansara and Lakshmi, 2021; Shamsudduha et al., 2017; Bonsor et al., 2018). Thus, a scale factor derived from land surface models was applied to the GRACE land water equivalent (LWE) thickness data by multiplying the raw GRACE LWE thickness by the scale factor. This is necessary to obtain a spatial resolution of 0.5° and it is not pre-applied to the GRC Tellus JPL Monthly Mass Grid Global Mascon data. Additionally, a gap in GRACE/GRACE-FO observations occurred in 2017–2018 due to mission lifespan and battery complications (Landerer et al., 2020). For purposes of this study, linear interpolation is used to fill the missing monthly TWSA data. These data were used to construct time series and spatial plots of monthly TWSA. Further information about the GRACE mission and the GRC Tellus JPL Monthly Mass Grid Global Mascon dataset can be found at https://gracefo.jpl.nasa.gov and https://podaac-tools.jpl.nasa.gov/drive/files/allData/gracefo/docs/GRACE-FO_L3_Handbook_JPL.pdf.

3. Methodology

3.1. *Hydrological components of the water balance equation*

The water balance equation is as follows:

$$P - ET - R = \Delta S \tag{1}$$

where P is precipitation, ET is evapotranspiration, R is runoff, and ΔS is change in total water storage. In this study, P and ET are derived from Earth observations (GPM IMERG and MODIS, respectively) while R is a model output (GLDAS). GRACE/GRACE-FO was used to estimate ΔS. Assessment of water balance closure in the Lake Victoria Basin using these datasets are limited by the MODIS ET observations because this dataset masks out lakes. Given the large spatial extent of Lake Victoria which makes up approximately 25% of the basin, it is unreasonable to assess water balance closure of the basin without evapotranspiration data over the lake itself. Thus, in this study we have chosen to examine individual hydrological components of the water balance equation (specifically precipitation, evapotranspiration over land, runoff, and change in water storage). Additional hydrological variables examined in this study include soil moisture, lake level height, and lake extent.

3.2. *Hydrological anomalies*

Monthly climate anomalies for precipitation, runoff, evapotranspiration, and root zone soil moisture were calculated using the difference between the monthly value and the monthly climatology. *Monthly climate anomaly* (henceforth referred to as 'anomaly') represents the difference between the *actual monthly value* and the *monthly climatology*. For example, a positive anomaly represents more rainfall than the climatological average. Likewise, a negative anomaly represents less rainfall than the climatological average. It is important to note that by international standards, climatology calculations should use a record of at least 30 years (WMO, 2017). However, this was not feasible for our study due to the availability and mission durations of Earth observations used for the analysis. Thus, a limitation of our study is that only a record of 20 years (June 2002 – June 2022) was used to calculate monthly climatology. Ideally, an equal time span of observations should be used for each month when calculating monthly climatology, but since the number of years of available observations is below the 30-year threshold, all available data is used for the analysis. Thus, 20 years of observations were used for all months except June, which had 21 years of observations.

Time series and spatial plots of monthly climate anomalies were constructed for each hydrological variable. For visual purposes,

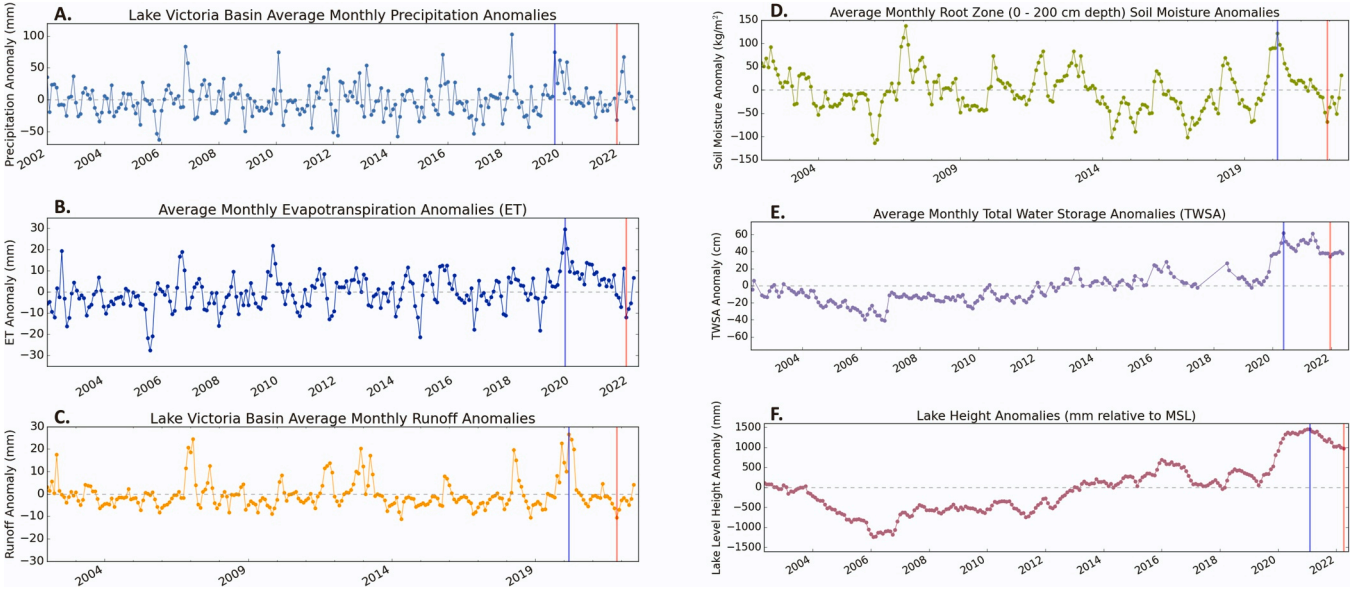


Fig. 3. Time series of average monthly anomalies for hydrological variables (precipitation, evapotranspiration, runoff, root zone soil moisture, total water storage anomaly (TWSA), and lake height) in the Lake Victoria Basin. The local maxima positive anomaly during the 2019–2020 flood period is indicated by a blue vertical line (representing the wet month). The local minima negative anomaly during the 2021–2022 drought period is indicated by a red vertical line (representing the dry month). Spatial anomaly plots of the indicated wet and dry months for each hydrological variable are shown in Fig. 4. (A) Time series of average monthly precipitation anomalies from Jan. 2002 – Jun. 2022 derived from the Integrated Multi-satellitE Retrievals for Global Precipitation Measurement (GPM IMERG). (B) Time series of average monthly evapotranspiration anomalies from Jun. 2002 – Jun. 2022 derived from the Moderate Resolution Imaging Spectroradiometer aboard the Terra mission. (C) Time series of average monthly runoff anomalies from Jun. 2002 – Jun. 2022 derived from the Global Land Data Assimilation System (GLDAS). (D) Time series of average monthly root zone soil moisture (0 – 2 m depth) anomalies from Jun. 2002 – Jun. 2022 derived from the Global Land Data Assimilation System (GLDAS). (E) Time series of total water storage anomaly (TWSA) from Jun. 2002 – May. 2022 derived from the Gravity Recovery and Climate Experiment (GRACE) mission. (F) Time series of average monthly lake height anomalies relative to mean sea level (MSL) derived from Jason-1, Jason-2, and Jason-3.

linear interpolation was used to re-grid precipitation, evapotranspiration, runoff, and root zone soil moisture to the same spatial scales for the spatial anomaly plots. TWSA was not re-gridded as this would be an overextension of the coarse native spatial resolution.

4. Results and discussion

4.1. Time series of hydrological anomalies

Time series of hydrological anomalies for precipitation, evapotranspiration, runoff, root zone soil moisture, TWSA, and lake height from June 2002 – June 2022 are shown in Fig. 3A–E. A key objective of this study is to examine the recent flood and drought periods in the Lake Victoria Basin, so our results and discussion are focused on hydrological anomalies between September 2019 – June 2022. However, the extended period of record provides valuable context for the magnitude and duration of the recent flood and drought events. The highest positive anomalies within this time period of June 2002 – June 2022 were observed during the 2020–2021 flood event for evapotranspiration, runoff, TWSA, and lake height. Negative anomalies during the 2021–2022 drought event however were not the most extreme negative anomalies that the Lake Victoria Basin has experienced in the past 20 years. In Fig. 3, the local maxima and minima hydrological anomalies are respectively indicated by vertical blue and red lines which show months where the Lake Victoria Basin experienced anomalous wet and dry conditions. Table 3 summarizes months within the period of interest that were identified as positive anomalies (wet month) and the negative anomalies (dry month), as well as their respective anomaly values where applicable. Spatial plots corresponding to these identified wet and dry months for precipitation, evapotranspiration, root zone soil moisture, and TWSA are discussed in Section 4.2.

In the precipitation anomalies time series shown in Fig. 3A, October 2019 was identified as the anomalous wet month with a value of + 73 mm. November 2021 was identified as the anomalous dry month with a value of − 32 mm. While the positive precipitation anomaly during October 2019 was not the most extreme observed within the basin between 2002 and 2022, it was followed by 8 months of persistently positive precipitation anomalies from September 2019 – April 2020, which is one of the longest periods of positive precipitation anomalies experienced in the basin.

The increase in TWSA during the same period (shown in Fig. 3E) concurs with the precipitation anomaly time series. TWSA reached its local maximum positive anomaly of + 62 cm in May 2020, which is one month later than the precipitation anomaly peak. This was the highest TWSA value observed within the Lake Victoria Basin since 2002. During the 2021–2022 drought event, TWSA remains positive. In the past however, the Lake Victoria Basin has experienced dry periods with strong negative anomalies which indicates the relative magnitude of the early 2022 drought in comparison to previous droughts.

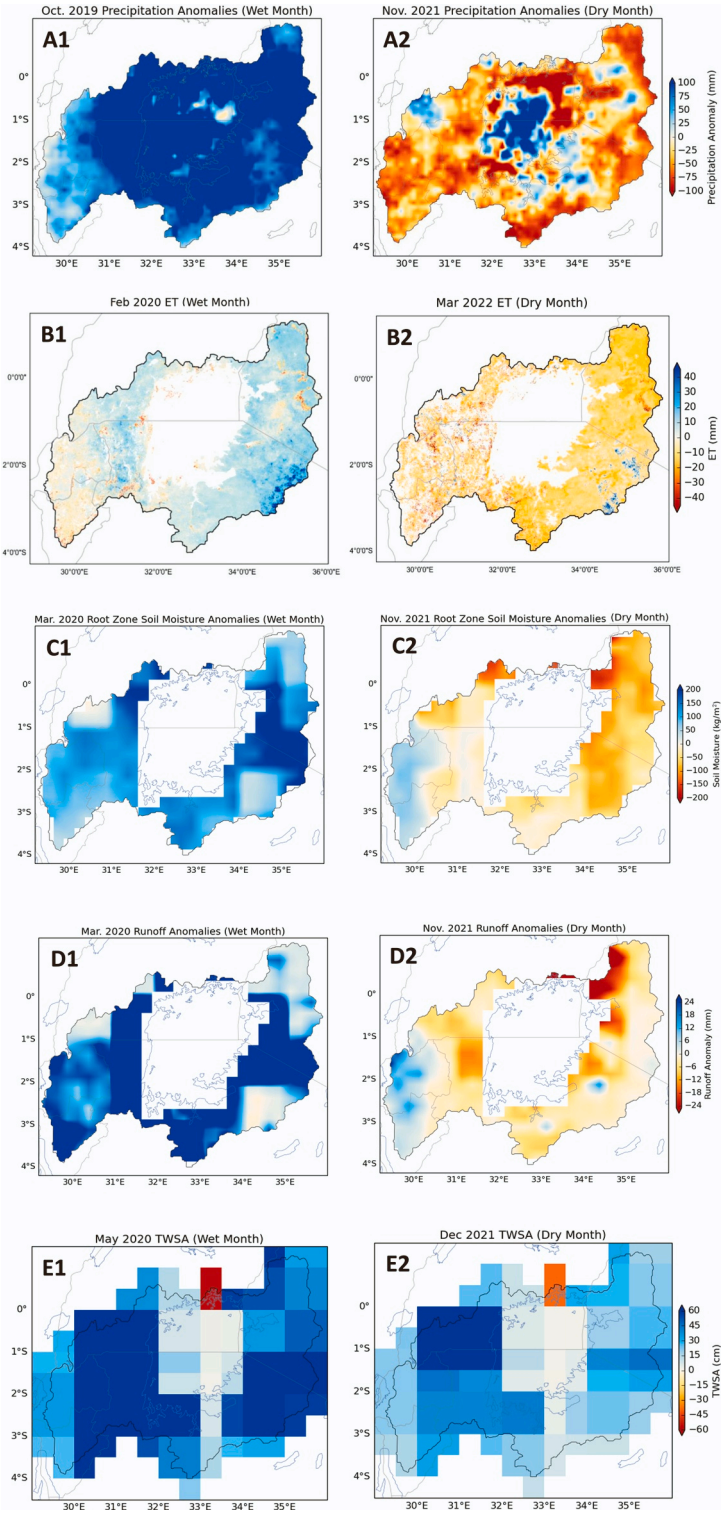
Lake height anomalies (shown in Fig. 3F) also increased substantially during the flood event. In September 2019, the lake height anomaly was + 340 mm. By May 2020, the lake height anomaly increased to + 1380 mm. At the time, lake level heights in May 2020 were record-breaking (BBC, 2020). However, the lake height proceeded to reach a higher positive anomaly in February 2021 with a value of + 1470 mm. During the 2021–2022 drought period, lake height anomalies remained positive while decreasing to + 972 mm. The TWSA in late 2021 and early 2022 remained relatively constant, despite the steady decrease in lake height anomalies at this time.

Root zone soil moisture anomalies (shown in Fig. 3D) fluctuate from positive to negative between the 2019–2020 flood and 2021–2022 drought events. A maximum root zone soil moisture anomaly of + 120 kg/m^2 was observed in March 2020. A minimum root zone soil moisture anomaly of − 71 kg/m^2 was observed in November 2021, which was then followed by five months of persistently negative root zone soil moisture anomalies. By June 2022, however, the root zone soil moisture anomalies recovered and were positive. This captures the quick recovery of the recent drought event.

Table 3
Summary of wet/dry months and positive/negative anomaly values for hydrological variables during the recent flood and drought events. These were determined from the time series of monthly basin-averaged anomalies constructed from 20 years of Earth observations or model data (see Fig. 3).

Hydrological Variable	Positive Anomaly (Wet Month)	Negative Anomaly (Dry Month)	Spatial Plot
Precipitation	Oct. 2019 + 73 mm	Nov. 2021 -32 mm	Yes
Evapotranspiration	Feb. 2020 + 29 mm	Mar. 2022 -12 mm	Yes
Runoff	Mar. 2020 + 27 mm	Nov. 2021 -10 mm	Yes
Root Zone Soil Moisture	Mar. 2020 + 120 kg/m^2	Nov. 2021 -71 kg/m^2	Yes
Near-surface Soil Moisture: 9-km*	Mar. 2020 n/a	Nov. 2021 n/a	Yes
Near-surface Soil Moisture: 1-km*	Mar. 2020 n/a	Nov. 2021 n/a	Yes
Lake Height	May 2020 + 1481 mm	Feb. 2022 + 972 mm	No
Total Water Storage Anomaly (TWSA)	May 2020 + 62 cm	Dec. 2021 + 36 cm	Yes

*: The wet and dry months for near-surface soil moisture were assumed to be the same as those identified by the root zone soil moisture anomaly time series, since there is not a sufficient observation record of near-surface soil moisture data to independently calculate climatology.



(caption on next page)

Fig. 4. Average monthly spatial anomalies of hydrological parameters during the anomalously high (wet) and low (dry) months between Sept. 2019 – Jan. 2022. These months are the local maxima and minima in the basin-averaged monthly time series of hydrological anomalies shown in Fig. 3 as the vertical blue and red lines to indicate wet and dry anomalies. For visual purposes, the scalebar was manually selected to represent variability within the basin by displaying outliers as the maximum/minimum colors. (A1) Monthly precipitation anomalies in Oct. 2019 (wet month). (A2) Monthly precipitation anomalies in Nov. 2021 (dry month). (B1) Monthly evapotranspiration anomalies in Mar. 2020 (wet month). (B2) Monthly evapotranspiration anomalies in Jan. 2022 (dry month). (C1) Monthly runoff anomalies in Mar. 2020 (wet month). (C2) Monthly runoff anomalies in Nov. 2021 (dry month). (D1) Monthly root zone soil moisture anomalies in Mar. 2020 (wet month). (D2) Monthly root zone soil moisture anomalies in Dec. 2021 (dry month). (E1) Monthly total water storage anomaly (TWSA) in May 2020 (wet month). (E2) Monthly total water storage anomaly (TWSA) in Dec. 2021 (dry month).

Similar to the root zone soil moisture anomalies, runoff anomalies (shown in Fig. 3C) fluctuate from positive values in March 2020 of + 27 mm to negative values in November 2021 of − 10 mm. The runoff anomalies are persistently negative from June 2021-May 2022, before recovering to a positive anomaly in June 2022. In conjunction with the root zone soil moisture anomalies during June 2022, the runoff anomalies in June 2022 also captures the basin's quick recovery from the recent drought event.

4.2. Spatial variability of hydrological anomalies

Spatial anomaly plots of precipitation, evapotranspiration, runoff, root zone soil moisture, and TWSA are shown in Fig. 4A-E. For each hydrological variable, the previously identified wet and dry months from the hydrological anomalies time series are spatially plotted. Scalebar ranges were manually selected to ensure average variabilities over the basin are visually apparent by displaying outliers as the maximum and minimum colors.

Precipitation anomalies during October 2019 (wet month) and November 2021 (dry month) are shown in Fig. 4A1-2. During the wet month, positive precipitation anomalies were dominant across the basin with most regions experiencing anomalies of + 100 mm or greater. This concurs with media and journal articles which reported heavy and atypical rainfall in the Eastern Horn of Africa due to one of the strongest Indian Ocean Dipole (IOD) events on record (Wainwright et al., 2020; FEWS NET, 2020). The greatest precipitation anomaly of + 472 mm was observed directly over Lake Victoria. In the dry month, precipitation anomalies were as severe as approximately − 160 mm over the northeastern and southwestern shores of Lake Victoria. A few patches of positive anomalies were observed throughout the basin such as directly over Lake Victoria. This is likely explained by local meteorology in the nocturnal lake-land breeze system of Lake Victoria which makes precipitation directly over the lake 25 – 30% greater than precipitation over the surrounding land (Shamsudduha et al., 2017; Thiery et al., 2016, 2017).

Evapotranspiration spatial anomaly plots for February 2020 (wet month) and March 2022 (dry month) are shown in Fig. 4B1–2. As previously discussed, an unfortunate limitation of this MODIS evapotranspiration product is that lakes and permanent waterbodies are masked out. However, one would expect that a significant portions of the basin's overall evapotranspiration occurs directly over Lake Victoria due to the lake's large surface area, shallow depth, and the region's tropical climate. Previous studies have found that evaporation accounts for 80% of the water leaving Lake Victoria while precipitation accounts for approximately 86% of the total water input into Lake Victoria (Nyeko-Ogiramoi et al., 2013). However, it is not possible to assess these results with the public Earth observations utilized in this study. In Fig. 4B1–2, less extreme anomalies (± 10 mm) were observed in the western portion of the basin compared to the eastern regions (± 40 mm). When compared alongside land cover type (Fig. 2), permanent wetlands are highlighted as positive anomalies during the wet month and the Kenyan forested areas are highlighted as negative anomalies during the dry month.

Similar spatial anomaly patterns were observed for runoff (Fig. 4C1–2) and root zone soil moisture (Fig. 4D1–2). During the wet month (March 2020), the most extreme runoff and root zone soil moisture anomalies were observed in the lower elevations (up to +120 mm for runoff and +254 kg/m^2 for root zone soil moisture anomalies). In the dry month (November 2021), root zone soil moisture anomalies ≤ -100 kg/m^2 were observed across most of the eastern portion of the basin. Additionally, there is a clear division between positive/negative runoff and root zone soil moisture anomalies in Burundi and Rwanda. This pattern follows the boundary of cropland versus grassland/savanna land cover types shown in Fig. 2.

TWSA spatial plots for May 2020 (wet month) and December 2021 (dry month) are shown in Fig. 4E1–2. For both periods, an overall positive TWSA was observed within the basin with regions experiencing peak TWSA greater than or equal to + 60 cm. However, December 2021 has lower positive anomalies than May 2020 in the eastern and southwestern portions of the basin. Observed changes in TWSA are more extreme over land as opposed to the lake itself between the wet and dry months. Additionally, highly negative TWSA are observed in Uganda near Lake Victoria's outlet which is a highly populated urban area. Because the native resolution of GRACE is coarse, it is possible that these observed patters are indicative of limitations of the enhanced spatial resolution of the GRC Tellus Global Mascon Monthly Mass Grid data. Further research should be conducted to assess the accuracy and limitations of this data product with regard to extreme hydrological events in the Lake Victoria Basin.

Overall, the spatial plots of hydrological anomalies capture significant variability within the Lake Victoria Basin and between periods of flood and drought. This provides valuable spatial hydrological information which complements the previously discussed temporal anomaly analysis. This level of spatial hydrological analysis would not otherwise be obtainable given the region's limited in-situ hydrological observations.

4.3. Comparison of 9-km and 1-km soil moisture products

Spatial plots of monthly soil moisture observations (0–5 m depth) are shown in Fig. 5. Due to the limited mission duration of SMAP (launched in 2015), a sufficient period of record was not available to conduct an anomaly analysis. Additionally, no soil moisture data is collected over Lake Victoria or other waterbodies within the basin.

Consistent with the root zone soil moisture anomaly analysis, March 2020 and November 2021 were selected as wet and dry months of interest. The 9-km SMAP soil moisture observations for the wet month show higher soil moisture values, ranging from approximately 0.34 to 0.44 kg/m^2 in the northeastern, southern, and southwestern regions of the basin (Fig. 5A1). During the dry month, the 9-km SMAP soil moisture observations showed that the lower elevations in the northeast maintained higher soil moisture values than the higher elevations in Kenya. The southeastern regions of the basin in Rwanda and Burundi also maintained higher soil moisture than the surrounding regions. Based on the land cover classifications, this spatial pattern may be explained by human intervention to maintain soil moisture conditions suitable for agriculture.

The 1-km downscaled soil moisture observations are shown in Fig. 5B1–2 for March 2020 and November 2021 as well. Overall, the 1-km soil moisture product demonstrated higher soil moisture values in the southern region compared to the 9-km SMAP product (Fig. 5B1). Similarly, in the dry month, the 1-km soil moisture product demonstrated lower soil moisture values (0.05 kg/m^2) than the 9-km SMAP product (0.13 kg/m^2) in the eastern edge of the basin in Kenya and Tanzania (Fig. 5B2). Qualitatively, the 1-km soil moisture product seems to capture more extreme soil moisture values than the 9-km SMAP product.

A quantitative validation analysis was performed with 28 ground-based soil moisture stations through the Trans-African Hydro-Meteorological Observatory (TAHMO) and the International Soil Moisture Network (ISMN) which is accessible at https://ismn.geo.tuwien.ac.at/en/. Validation statistics, plots, and time series are shown in Figs. 6 and 7 and the Appendix (1–2). It is important to

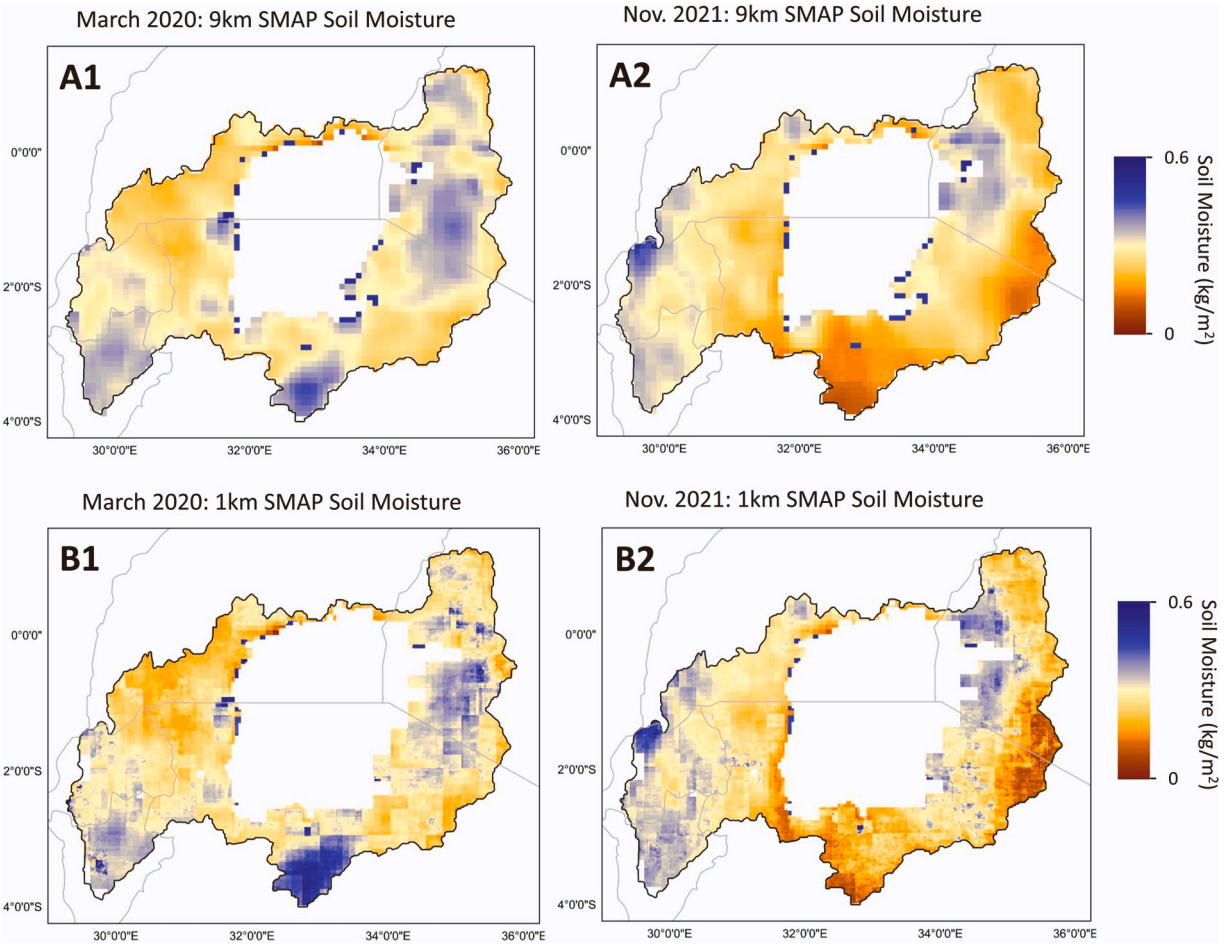


Fig. 5. Spatial plots of monthly soil moisture (0–5 cm depth) observations derived from SMAP. March 2020 and November 2021 were respectively selected as the wet and dry months, based on the time series of monthly root zone soil moisture anomalies (Fig. 4 C1 and C2). The spatial plots are shown at both 9-km and 1-km resolutions. (A1) 9-km monthly soil observations in Mar. 2020 (wet month); (A2) 9-km monthly soil moisture observations in Nov. 2021 (dry month); (B1) 1-km monthly soil moisture observations in Mar. 2020 (wet month); (B2) 1-km monthly soil moisture observations in Nov. 2021 (dry month).

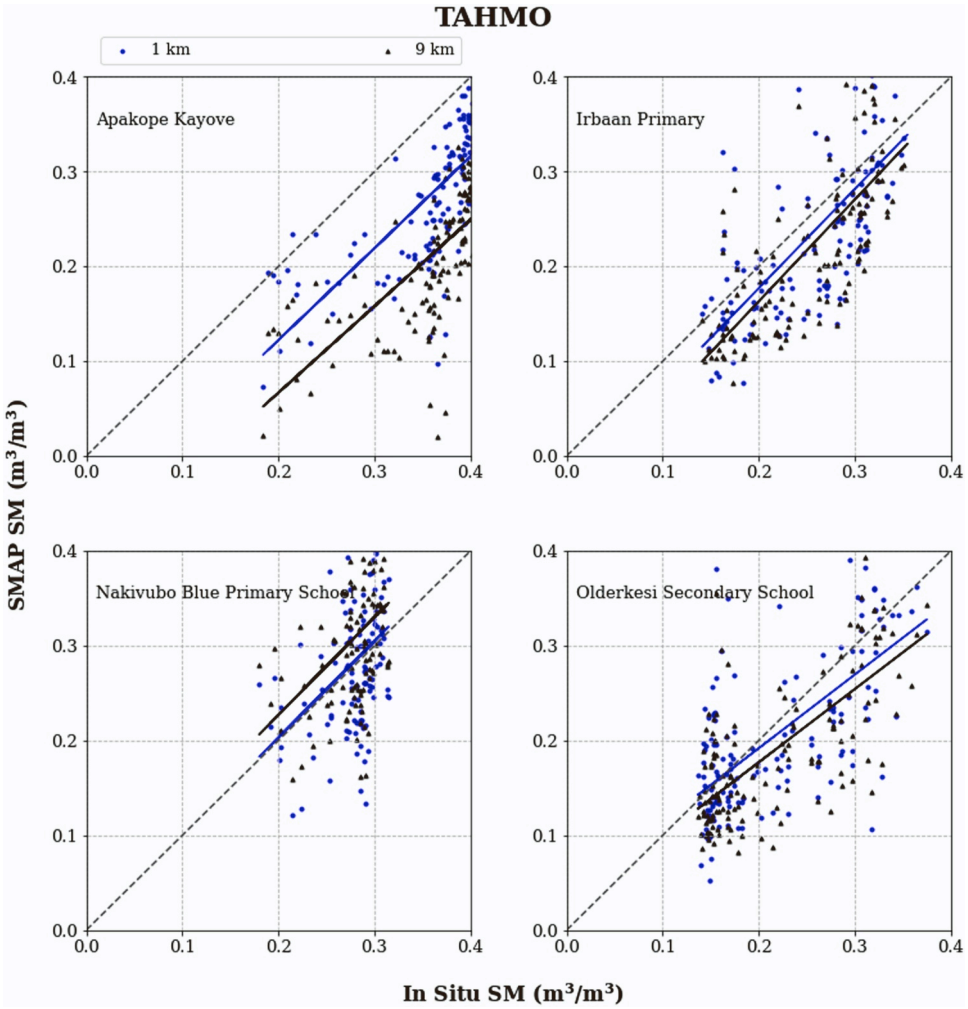


Fig. 6. Results of validation of in-situ near-surface soil moisture observations from the International Soil Moisture Network with the 1-km and 9-km Soil Moisture Active Passive (SMAP) observations at 6:00AM overpasses.

note that this validation is limited by the short period of in-situ observations primarily from mid-2018 to present. Additionally, no ISMN in-situ observations were available in Tanzania.

In Fig. 6, data collected at four ISMN soil moisture observation sites were validated with the 1-km and 9-km SMAP observations. The 1-km downscaled soil moisture observations performed better than the 9-km soil moisture product for each site. Fig. 7 shows time series of the same four ISMN soil moisture observation sites in comparison with the 1-km and 9-km soil moisture SMAP observations from 2015 to 2021. Here, it is possible to see how the performance of both the 1-km and 9-km soil moisture products vary over time in comparison with the in-situ soil moisture observations. In general, the 1-km data performed comparably or better than the 9-km SMAP observations with lower unbiased RMSE. Both the 1-km and 9-km data had low p-values and were statistically significant. From this validation (Figs. 6 and 7) and the Appendix, it is clear that the downscaled 1-km soil moisture compares better to the available in-situ observations than the original 9-km soil moisture data. Along with the 1-km global validation conducted by Fang et al. (2022), this validation forms a good basis for acceptance of use of the SMAP downscaled 1-km soil moisture data set for analyses in this basin. Future research should be conducted to continue validation efforts across the African continent.

4.4. Change in lake volume

To further quantify the relative hydrological variations in this region between hydrological extremes, change in the volume of Lake Victoria were calculated from the identified wet month (May 2021) and dry month (Feb. 2022). The following equation was used:

$$\Delta V = \Delta A * \Delta H$$

where ΔV is change in lake volume, ΔA is change in lake surface area, and ΔH is change in lake level height. The change in surface area

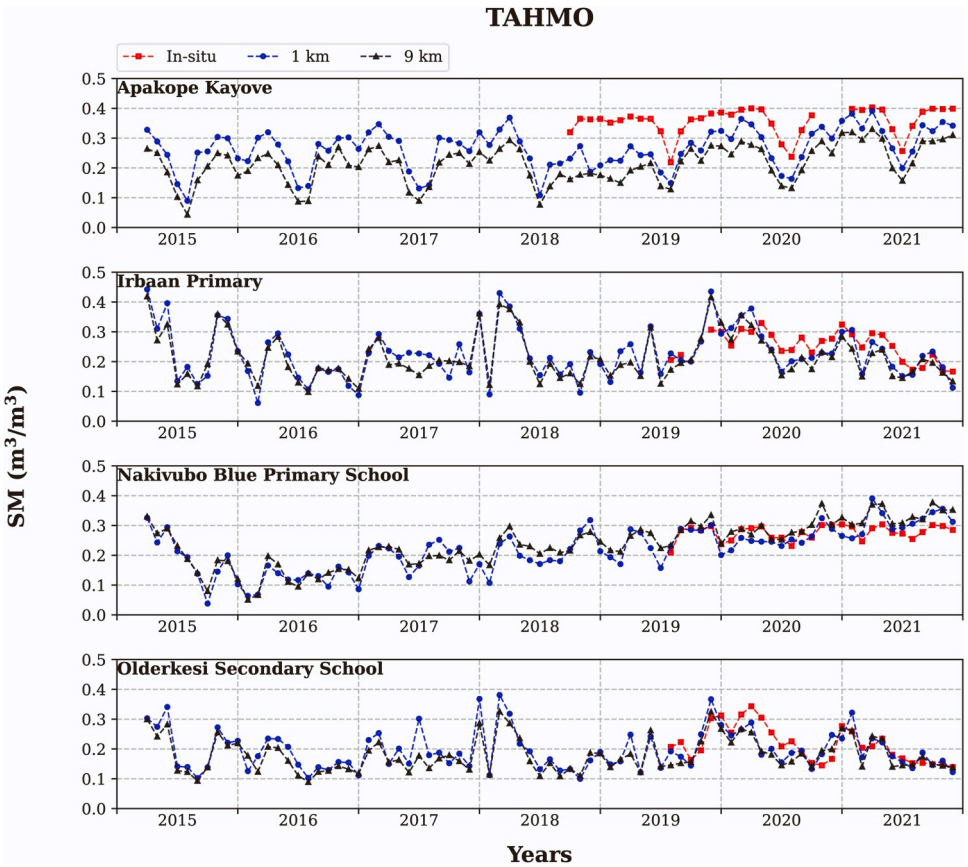


Fig. 7. Time series of in-situ near-surface soil moisture observations from four TAHMO stations compared with 1-km and 9-km SMAP soil moisture observations from 2015 to 2021.

between May 2021 and Feb. 2022 were calculated by identifying the total area of inundated pixels over Lake Victoria in Sentinel-1 synthetic aperture radar (SAR) imagery during the two months. Changes in lake level height were derived from the G-REALM altimetry data.

Between May 2021 and Feb. 2022, the lake level height decreased by 1.19 m and the surface area decreased by approximately 9270 km^2. Thus, the overall change in lake level volume was − 11.03 km^3. This large decrease in lake volume concurs with the results derived from the other hydrological variables which also experienced large decreases (represented as negative anomalies) during this

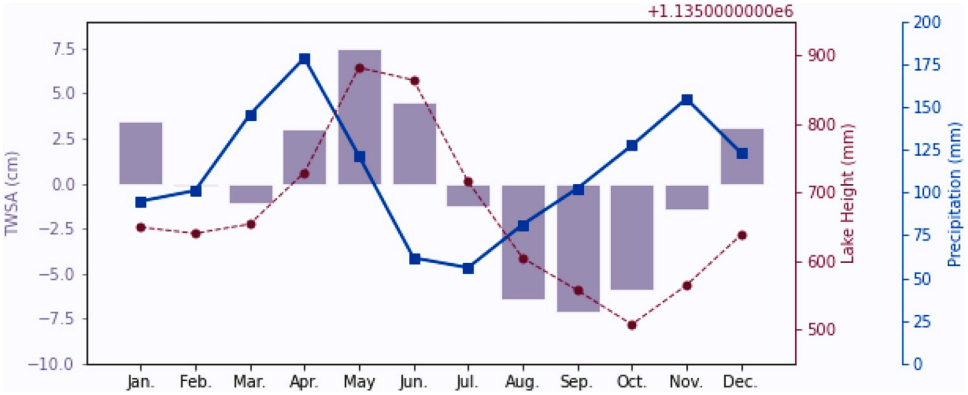


Fig. 8. Variability of monthly climatological values of precipitation, lake height, and total water storage anomaly (TWSA) which were calculated from the duration of the study period from June 2002 – January 2022 using equation 2. This graph shows the seasonality of precipitation, lake height, and TWSA within the basin. It is important to note that lake height is managed by hydropower dams at Lake Victoria's outlet near Jinja, Uganda.

time period. This is apparent in the basin-wide averaged monthly TWSA observations which decreased from + 60 cm in May 2021 to + 35 cm in Feb. 2022. The large decreases in lake volume and TWSA during this period further indicate the occurrence of these two hydrological extreme events.

4.5. Relationship between TWSA, precipitation, and lake height

A time series of monthly climatology for precipitation, lake height, and TWSA are shown in Fig. 8 (see Section 3.2, equation 2 for more details on the calculation of *monthly climatology*). This depicts the seasonality of precipitation, lake height, and TWSA based on the period of available data (June 2002 – June 2022 for precipitation and lake height; June 2002 – May 2022 for TWSA). The precipitation monthly climatology captures the distinct short rainy season and long rainy season which respectively occur between March – May and September – November. April has the highest average rainfall of 118 mm/month. July has the lowest average rainfall of 56 mm/month. Lake height follows similar seasonality as precipitation with a slight lag (approximately 1–2 months). May has the highest average lake height with an elevation of 1136 m above MSL. October has the lowest average lake height with an elevation of 1134 m above MSL. As previously discussed, Lake Victoria is sensitive to human management of hydropower dam operations at its outlet so these seasonal changes may be a result of both anthropogenic and natural factors. TWSA depicts similar seasonality as precipitation with a lag of 1–2 months. May has the highest average monthly TWSA value of + 7 cm/month. September has the lowest average monthly TWSA value of − 8 cm/month.

Fig. 9 shows a longer time series comparison from January 2019 – January 2022 for monthly TWSA (purple bars), observed basin-averaged monthly precipitation (blue solid line), and monthly climatology of precipitation (dashed gray line). From October 2019 to May 2020, the average observed monthly precipitation exceeded the monthly precipitation climatology by 13–75 mm. This means the basin experienced a positive precipitation anomaly and was wetter than the climatological average. From media articles, it is documented that extensive flooding occurred during this period in the Lake Victoria Basin and East Africa (BBC, 2020; FEWS NET, 2020; Kisumu, 2020; Khaki and Awange, 2021; Wainwright et al., 2021).

Similarly, March 2021 – December 2021 had average observed monthly precipitation below the monthly climatology by − 6 mm to − 31 mm. This indicates a negative precipitation anomalies and drier conditions than the climatological average. These observations concur with a report from the Famine Early Warning Systems Network which documented this period as having a failed rainy season and high risk for crop failure and food insecurity in East Africa (FEWS NET, 2021).

5. Conclusions

Over the previous decades, hydrometeorological extreme like floods and droughts have been continual and recurrent in the Lake Victoria Basin. Recently, the Lake Victoria Basin experienced extreme floods in 2019–2020 and drought conditions in early 2022. High-resolution spatial and temporal monitoring of hydrological variables are vital for improving understanding of hydrology within the Lake Victoria Basin, particularly given the region's dense population and vulnerability to sub-seasonal hydrological anomalies which can lead to the loss of life and property, crop failure, food insecurity, and increased poverty. However, there are insufficient in-situ hydrological observations in the Lake Victoria Basin to fill this need. Thus, in this study, we leverage publicly available Earth observations and model data to capture occurrence of the recent flood and drought events in the Lake Victoria Basin. We examined and discussed components of the hydrological water balance (precipitation, evapotranspiration, runoff, and change in total water storage) and additional hydrological variables (soil moisture, lake height, and lake extent) with specific regard to the recent hydrological extreme events. This was accomplished through construction of time series and spatial plots of monthly anomalies for each variable. Our results improve understanding and documentation of the severity and duration of the recent flood and drought events.

The innovation of this work is that it offers a simple method of using publicly available Earth observations and model data to improve spatial and temporal analyses of hydrological anomalies at sub-seasonal time scales within a region lacking in-situ data. To our knowledge, this is the first study to conduct a hydrological spatial anomaly comparison for the 2019–2020 flood and 2021–2022

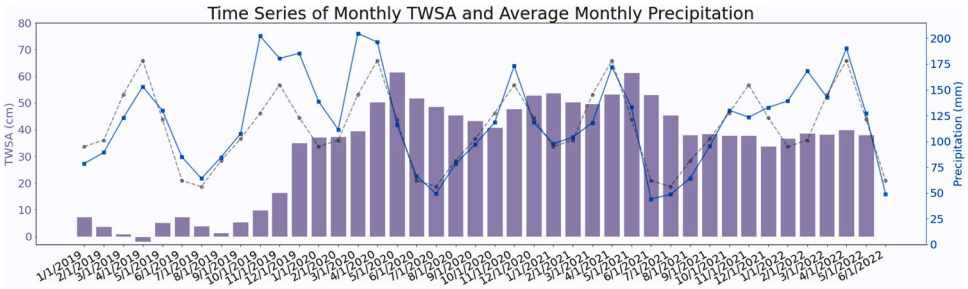


Fig. 9. Time series of monthly total water storage anomaly (TWSA, shown as purple bars), observed basin-averaged monthly precipitation (shown as solid blue line), and climatological monthly precipitation (black dashed line) from Jan. 2019 – Jan. 2022. If the monthly precipitation is greater than the climatological precipitation, the basin experienced a positive anomaly meaning it was wetter than the climatological average. Likewise, if the monthly precipitation is less than the climatological precipitation, the basin experienced a negative anomaly meaning it was drier than the climatological average. Note, TWSA data was not yet published for Nov. 2021-Jan. 2022 when this analysis was conducted.

drought events in the Lake Victoria Basin. Thus, this is a novel contribution to the existing record of hydrological research in this region because it updates previous studies and is among the first to integrate such a diverse array of publicly available datasets for the Lake Victoria Basin. Further work could assess this method for informing water resources management in this region, particularly for informing scheduling, planting, irrigation, and harvesting of crops. This would be valuable given the Lake Victoria Basin's growing population and economic reliance on rain-fed subsistence agriculture.

Additionally, a comparison of 9-km and 1-km near-surface soil moisture products derived and downscaled from SMAP was conducted to improve spatial resolution of soil moisture observations in the Lake Victoria Basin. The 1-km near-surface soil moisture product was validated with ground station near-surface soil moisture data available through the International Soil Moisture Network and performed either better or comparably to the 9-km SMAP observations. This was the first validation of the 1-km near-surface soil moisture product in the continent of Africa. Further research should continue validation efforts of soil moisture data derived from Earth observations in the Lake Victoria Basin and Africa in the limited regions where ground-based observations are available.

Methods and findings from this study can be used to advance understanding and monitoring methods of sub-seasonal hydrological anomalies within the Lake Victoria Basin, particularly through publicly available Earth observations. By demonstrating a simple method for improved monitoring of hydrological variables including precipitation, evapotranspiration, runoff, soil moisture, lake height, and TWSA, this study supports the continual effort to decrease the Lake Victoria Basin community's vulnerability to hydrological anomalies during extreme floods and droughts.

CRediT authorship contribution statement

We certify that Gigi Pavur and Venkataraman Lakshmi contributed to the content, concept, design, analysis, writing, and revision of this manuscript, "Observing the Recent Floods and Drought in the Lake Victoria Basin Using Earth Observations and Hydrological Anomalies."

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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Appendix 1

Summary of four soil moisture stations used for validation from the Trans-African Hydro-Meteorological Observatory (TAHMO) network. Data was accessed through the International Soil Moisture Network (ISMN).

ISMN Station Name	Country	Observation Depth (m)	Start Date	End Date
Apakope Kayove	Rwanda	0.20	10/23/2018	12/31/2021
Irbaan Primary	Kenya	0.20	8/2/2019	12/31/2021
Nakivubo Blue Primary School	Uganda	0.20	8/22/2019	12/31/2021
Olderkesi Secondary School	Kenya	0.2	8/6/2019	12/31/2021

Appendix 2

Metrics and statistics from validation of in-situ near-surface soil moisture observations from the International Soil Moisture Network (ISMN) with the 1-km and 9-km Soil Moisture Active Passive (SMAP) observations at 6:00AM overpasses. Number of observations, R^2, unbiased root mean square error (ubRMSE), bias, and p-value are provided for 1-km (table A) and 9-km (table B) SMAP validation at each site.

A: Validation Statistics of 1-km SMAP Observations

ISMN Station Name	Number of Observations	R^2	ubRMSE	Bias	p-value
Apakope Kayove	126	0.494085	0.082586	0.082567	4.54E-20
Irbaan Primary	130	0.505022	0.020658	0.020462	2.79E-21
Nakivubo Blue Primary School	111	0.153645	0.004623	-0.00459	2.1E-05
Olderkesi Secondary School	141	0.427319	0.019615	0.011919	1.53E-18

B: Validation Statistics of 9-km SMAP Observations

ISMN Station Name	Number of Observations	R^2	ubRMSE	Bias	p-value
Apakope Kayove	126	0.457544	0.146901	0.146819	3.55E-18
Irbaan Primary	130	0.548914	0.033146	0.032838	7.04E-24
Nakivubo Blue Primary School	111	0.159815	0.029315	-0.0293	1.38E-05
Olderkesi Secondary School	141	0.477956	0.031178	0.026849	2.33E-21

References

Agutu, N.O., Awange, J.L., Zerihun, A., Ndehedehe, C.E., Kuhn, M., Fukuda, Y., 2017. Assessing multi-satellite remote sensing, reanalysis, and land surface models' products in characterizing agricultural drought in East Africa. Remote Sens. Environ. 194, 287–302. https://doi.org/10.1016/j.rse.2017.03.041.

Agutu, N.O., Ndehedehe, C.E., Awange, J.L., Kirimi, F., Mwaniki, M., 2021. Understanding uncertainty of model-reanalysis soil moisture within Greater Horn of Africa (1982 - 2014). J. Hydrol. 603. https://doi.org/10.1016/j.jhdrol.2021.127169.

Alsdorf, D., Beighley, E., Laraque, A., Lee, H., Tshimanga, R., O'Loughlin, F., Mahe, G., Dinga, B., Moukandi, G., Spencer, R.G., 2016. Opportunities for hydrologic research in the Congo Basin. Rev. Geophys. 54 (2), 378–409. https://doi.org/10.1002/2016RG000517.

Awange, J.L., 2006. Lake Victoria: Ecology, Resources, Environment. Springer Science & Business Media.

Awange, J.L., 2020. Lake Victoria Monitored from Space. Springer Nature.

Awange, J.L., Sharifi, M.A., Ogonda, G., Wickert, J., Grafarend, E.W., Omulo, M.A., 2008. The falling Lake Victoria water level: GRACE, TRIMM and CHAMP satellite analysis of the lake basin. Water Resour. Manag. 22 (7), 775–796. https://doi.org/10.1007/s11269-007, 9191-y.

Awange, J.L., Hu, K.X., Khaki, M., 2019. The newly merged satellite remotely sensed, gauge and reanalysis-based multi-source weighted-ensemble precipitation: evaluation over Australia and Africa (1981–2016). Sci. Total Environ. 670, 448–465. https://doi.org/10.1016/j.scitotenv.2019.03.148.

BBC News, 2020. Ugandan Floating Island Causes National Power Cut. BBC. Available: ⟨https://www.bbc.com/news/world-africa-52286296⟩.

Becker, M., LLovel, W., Cazenave, A., Güntner, A., Crétaux, J.F., 2010. Recent hydrological behavior of the East African great lakes region inferred from GRACE, satellite altimetry and rainfall observations. Comptes Rendus Geosci. 342 (3), 223–233. https://doi.org/10.1016/j.crte.2009.12.010.

Birkett, M, C., Beckley, B., 2010. Investigating the performance of the Jason-2/OSTM radar altimeter over lakes and reservoirs. Mar. Geod. 33 (S1), 204–238. https://doi.org/10.1080/01490419.2010.488983.

Bonsor, H.C., Shamsudduha, M., Marchant, B.P., MacDonald, A.M., Taylor, R.G., 2018. Seasonal and decadal groundwater changes in african sedimentary aquifers estimated using GRACE products and LSMs. Remote Sens. 10 (6), 904. https://doi.org/10.3390/rs10060904.

Chan, S.K., Bindlish, R., O'Neill, P.E., Njoku, E., Jackson, T., Colliander, A., Chen, F., Burgin, M., Dunbar, S., Piepmeier, J., Yueh, S., Entekhabi, E., Cosh, M.H., Caldwell, T., Walker, J., Wu, X., Berg, A., Rowlandson, T., Pacheco, A., McNairn, H., Thibeault, M., Martínez-Fernández, J., González-Zamora, Á., Seyfried, M., Bosch, D., Starks, P., Goodrich, D., Prueger, J., Palecki, M., Small, E.E., Zreda, M., Calvet, J., Crow, W.T., Kerr, Y., 2016. Assessment of the SMAP passive soil moisture product. IEEE Trans. Geosci. Remote Sens. 54 (8), 4994–5007. https://doi.org/10.1109/TGRS.2016.2561938.

Coughlan de Perez, E., van Aalst, M., Choularton, R., van den Hurk, B., Mason, S., Nissan, H., Schwager, S., 2019. From rain to famine: assessing the utility of rainfall observations and seasonal forecasts to anticipate food insecurity in East Africa. Food Secur. 11 (1), 57–68. https://doi.org/10.1007/s12571-018-00885-9.

Dezfuli, A.K., Ichoku, C.M., Huffman, G.J., Mohr, K.I., Selker, J.S., Van De Giesen, N., Hochreutener, R., Annor, F.O., 2017. Validation of IMERG precipitation in Africa. J. Hydrometeorol. 18 (10), 2817–2825. https://doi.org/10.1175/JHM-D-17-0139.1.

Entekhabi, D., Njoku, E.G., O'Neill, P.E., Kellogg, K.H., Crow, W.T., Edelstein, W.N., Entin, J.K., Goodman, S.D., Jackson, T.J., Johnson, J., Kimball, J., Piepmeier, J. R., Koster, R.D., Martin, N., McDonald, K.C., Moghaddam, M., Moran, S., Reichle, R., Shi, J.C., Spencer, M.W., Thurman, S.W., Tsang, L., Van Zyl, J., 2010. The soil moisture active passive (SMAP) mission. Proc. IEEE 98 (5), 704–716. https://doi.org/10.1109/JPROC.2010.2043918.

Esri, 2021. DeLorme, CIA World Factbook United Nations Development Programme, Flagpedia. African Countries Feature Service. Esri Africa [Dataset].

Fang, B., Lakshmi, V., Cosh, M., Liu, P.W., Bindlish, R., Jackson, T.J., 2022. A global 1km downscaled SMAP soil moisture product based on thermal inertia theory. Vadose Zone J. 21 (2), e20182 https://doi.org/10.1002/vzj2.20182.

FEWS NET, 2020. East African Special Report: 2019 Short Rains in East Africa Among the Wettest on Historical Record. Famine Early Warning Systems Network. Available: ⟨https://fews.net/sites/default/files/documents/reports/East_Africa_Special_Report_January_2020_Final_3.pdf⟩.

FEWS NET, 2021. East African Food Security Alert: Over 20 million people in need of urgent food aid in the Horn of Africa amid severe drought and conflict. Famine Early Warning Systems Network. Available: ⟨https://reliefweb.int/sites/reliefweb.int/files/resources/east-africa-alert-20211229-final.pdf⟩.

Friedl, D.S., Sulla-Menashe, M.D., 2019. MCD12Q1 MODIS/Terra+Aqua Land Cover Type Yearly L3 Global 500m SIN Grid v006. NASA EOSDIS Land Processes DAAC [Dataset].

Getirana, A., Jung, H.C., Van Den Hoek, J., Ndehedehe, C.E., 2020. Hydropower dam operation strongly controls Lake Victoria's freshwater storage variability. Sci. Total Environ. 726, 138343 https://doi.org/10.1016/j.scitotenv.2020.138343.

Grippa, M., Kergoat, L., Frappart, F., Araud, Q., Boone, A., De Rosnay, P., Lemoine, J.M., Gascoin, S., Balsamo, G., Decharme, C.O.B., Saux-Picart, S., Ramillien, G., 2011. Land water storage variability over West Africa estimated by Gravity Recovery and Climate Experiment (GRACE) and land surface models. Water Resour. Res. 47 (5) https://doi.org/10.1029/2009WR008856.

Hamilton, S., 2016. Basin, Lake Victoria Watershed (inside), vector polygon, ~2015. Harvard Dataverse, V3, ⟨10.7910/DVN/Z5RMYD⟩.

Hashemi, H., Nordin, M., Lakshmi, V., Huffman, G.J., Knight, R., 2017. Bias correction of long-term satellite monthly precipitation product (TRMM 3B43) over the conterminous United States. J. Hydrometeorol. 18 (9), 2491–2509. https://doi.org/10.1175/JHM-D-17 -0025.1.

Hou, A.Y., Kakar, R.K., Neeck, S., Azarbarzin, A.A., Kummerow, C.D., Kojima, M., Oki, R., Nakamura, K., Iguchi, T., 2014. The global precipitation measurement mission. Bull. Am. Meteorol. Soc. 95 (5), 701–722. https://doi.org/10.1175/BAMS-D-13-00164.1.

Hu, K.X., Awange, J.L., Kuhn, M., Nanteza, J., 2021. Inference of the spatio-temporal variability and storage potential of groundwater in data-deficient regions through groundwater models and inversion of impact factors on groundwater, as exemplified by the Lake Victoria Basin. Sci. Total Environ. 800, 149355 https://doi.org/10.1016/j.scitotenv.2021.149355.

Huffman, G.J., Bolvin, D.T., Braithwaite, D., Hsu, K., Joyce, R., Xie, P., Yoo, S.H., 2015. NASA Global Precipitation Measurement (GPM) Integrated Multi-satellitE Retrievals for GPM (IMERG). Algorithm Theoretical Basis Document (ATBD) Version, 4, p. 26.

ICPAC, 2019. Lake Victoria Basin – Shapefile. Intergovernmental Authority on Development (IGAD) Climate Prediction and Applications Centre (IPAC). Available: ⟨http://geoportal.icpac.net/layers/geonode%3Alv_basin⟩ [Dataset].

Kaneda, T., Greenbaum, C., Kline, K., 2020. 2020 World Population Data Sheet. Population Reference Bureau. Available: ⟨https://bit.ly/3tJgT51⟩.

Kansara, P., Li, W., El-Askary, H., Lakshmi, V., Piechota, T., Struppa, D., Sayed, M.A., 2021. An assessment of the filling process of the Grand Ethiopian Renaissance Dam and its impact on the downstream countries. Remote Sens. 13 (4), 711. https://doi.org/10.3390/rs13040711.

Khaki, M., Awange, J., 2021. The 2019–2020 rise in Lake Victoria monitored from space: exploiting the state-of-the-art GRACE-FO and the newly released ERA-5 reanalysis products. Sensors 21 (13), 4304, 0.3390/s21134304.

Khan, S.I., Hong, Y., Wang, J., Yilmaz, K.K., Gourley, J.J., Adler, R.F., Brakenridge, G.R., Policelli, F., Habib, S., Irwin, D., 2010. Satellite remote sensing and hydrologic modeling for flood inundation mapping in Lake Victoria basin: implications for hydrologic prediction in ungauged basins. IEEE Trans. Geosci. Remote Sens. 49 (1), 85–95. https://doi.org/10.1109/TGRS.2010.2057513.

Kisumu, E.R., 2020. 'The Water Will Come Back': Why Kenya's Struggle Against Flooding Is Far from Over. The Guardian: Global Development. Available: ⟨https://www.theguardian.com/global-development/2020/jun/02/the-water-will-come-back-why-kenyas-struggle-against-flooding-is-far-from-over⟩.

Kull, D., 2006. Connections Between Recent Water Level Drops in Lake Victoria, Dam Operations and Drought. Available: ⟨https://aquadocs.org/handle/1834/7032⟩.

Lakshmi, V., Fayne, J., Bolten, J., 2018. A comparative study of available water in the major river basins of the world. J. Hydrol. 567, 510–532. https://doi.org/10.1016/j.jhydrol.2018.10.038.

Landerer, F.W., Swenson, S.C., 2012. Accuracy of scaled GRACE terrestrial water storage estimates. Water Resour. Res. 48 (12) https://doi.org/10.1029/2011WR011453.

Landerer, F.W., Flechtner, F.M., Save, H., Webb, F.H., Bandikova, T., Bertiger, W.I., Bettadpur, S.V., Byun, S.H., Dahle, C., Dobslaw, H., Fahnestock, E., Harvey, N., Kang, Z., Kruizinga, G.L.H., Loomis, B.D., McCullough, C., Murbock, M., Nagel, P., Paik, M., Pie, N., Poole, S., Strekalov, D., Tamisiea, M.E., Wang, F., Watkins, M. W., Wen, H.Y., Wiese, D.N., Yuan, D.N., 2020. Extending the global mass change data record: GRACE Follow-On instrument and science data performance. e2020GL088306 Geophys. Res. Lett. 47 (12). https://doi.org/10.1029/2020GL088306.

Le, H.M., Sutton, J.R., Bui, D.D., Bolten, J.D., Lakshmi, V., 2018. Comparison and bias correction of TMPA precipitation products over the lower part of red–Thai Binh River basin of Vietnam. Remote Sens. 10 (10), 1582. https://doi.org/10.3390/rs10101582.

Le, M.H., Lakshmi, V., Bolten, J., Du Bui, D., 2020. Adequacy of satellite-derived precipitation estimate for hydrological modeling in Vietnam Basins. J. Hydrol. 586, 124820 https://doi.org/10.1016/j.jhydrol.2020.124820.

Le, M.H., Nguyen, B.Q., Pham, H.T., Patil, A., Do, H.X., Ramsankaran, R.A.A.J., Bolton, J., Lakshmi, V., 2022. Assimilation of SMAP products for improving streamflow simulations over tropical climate region—is spatial information more important than temporal information? Remote Sens. 14 (7), 1607. https://doi.org/10.3390/rs14071607.

López-Carr, D., Pricope, N.G., Aukema, J.E., Jankowska, M.M., Funk, C., Husak, G., Michaelsen, J., 2014. A spatial analysis of population dynamics and climate change in Africa: potential vulnerability hot spots emerge where precipitation declines and demographic pressures coincide. Popul. Environ. 35 (3), 323–339. https://doi.org/10.1007/s11111-014-0209-0.

Mohammed, I.N., Bolten, J.D., Srinivasan, R., Lakshmi, V., 2018a. Improved hydrological decision support system for the Lower Mekong River Basin using satellite-based earth observations. Remote Sens. 10 (6), 885. https://doi.org/10.3390/rs10060885.

Mohammed, I.N., Bolten, J.D., Srinivasan, R., Lakshmi, V., 2018b. Satellite observations and modeling to understand the Lower Mekong River Basin streamflow variability. J. Hydrol. 564, 559–573. https://doi.org/10.1016/j.jhydrol.2018.07.030.

Mohammed, I.N., Bolten, J.D., Srinivasan, R., Meechaiya, C., Spruce, J.P., Lakshmi, V., 2018c. Ground and satellite based observation datasets for the Lower Mekong River Basin. Data Brief 21, 2020–2027. https://doi.org/10.1016/j.dib.2018.11.038.

Mondal, A., Lakshmi, V., Hashemi, H., 2018. Intercomparison of trend analysis of multisatellite monthly precipitation products and gauge measurements for river basins of India. J. Hydrol. 565, 779–790. https://doi.org/10.1016/j.jhydrol.2018.08.083.

Mu, Q., Heinsch, F.A., Zhao, M., Running, S.W., 2007. Development of a global evapotranspiration algorithm based on MODIS and global meteorology data. Remote Sens. Environ. 111 (4), 519–536. https://doi.org/10.1016/j.rse.2007.04.015.

Mu, Q., Zhao, M., Running, S.W., 2011. Improvements to a MODIS global terrestrial evapotranspiration algorithm. Remote Sens. Environ. 115 (8), 1781–1800. https://doi.org/10.1016/j.rse.2011.02.019.

Mugo, R., Waswa, R., Nyaga, J.W., Ndubi, A., Adams, E.C., Flores-Anderson, A.I., 2020. Quantifying land use land cover changes in the Lake Victoria basin using satellite remote sensing: the trends and drivers between 1985 and 2014. Remote Sens. 12 (17), 2829. https://doi.org/10.3390/rs12172829.

Ndehedehe, C., Awange, J., Agutu, N., Kuhn, M., Heck, B., 2016. Understanding changes in terrestrial water storage over West Africa between 2002 and 2014. Adv. Water Resour. 88, 211–230. https://doi.org/10.1016/j.advwatres.2015.12.009.

Nyeko-Ogiramoi, P., Willems, P., Ngirane-Katashaya, G., 2013. Trend and variability in observed hydrometeorological extremes in the Lake Victoria basin. J. Hydrol. 489, 56–73. https://doi.org/10.1016/j.jhydrol.2013.02.039.

Paris, A., Calmant, S., Gosset, M., Fleischmann, A.S., Conchy, T.S.X., Garambois, P.A., Bricquet, J.P., Papa, F., Tshimanga, R.M., Guzanga, G.G., Siqueria, V.A., Tondo, B.L., Paiva, R., Laraque, A., 2022. Monitoring hydrological variables from remote sensing and modeling in the Congo River Basin. Congo Basin Hydrology, Climate, and Biogeochemistry: A Foundation for the Future, pp. 339–366. https://doi.org/10.1002/9781119657002.ch18.

Ramillien, G., Frappart, F., Seoane, L., 2014. Application of the regional water mass variations from GRACE satellite gravimetry to large-scale water management in Africa. Remote Sens. 6 (8), 7379–7405. https://doi.org/10.3390/rs6087379.

Reynolds, C., 2005. Low Water Levels Observed on Lake Victoria. Production Estimates and Crop Assessment Division of USDA Foreign Agricultural Service. Available: ⟨https://reliefweb.int/report/kenya/usda-low-water-levels-observed-lake-victoria⟩.

Rodell, M., Houser, P.R., Jambor, U.E.A., Gottschalck, J., Mitchell, K., Meng, C.J., Arsenault, K., Cosgrove, B., Radakovich, J., Bosilovich, M., Entin, J.K., Walker, J.P., Lohmann, D., Toll, D., 2004. The global land data assimilation system. Bull. Am. Meteorol. Soc. 85 (3), 381–394. https://doi.org/10.1175/BAMS-85-3-381.

Running, S.W., Mu, Q., Zhao, M., Moreno, A., 2017. MODIS Global Terrestrial Evapotranspiration (ET) Product (NASA MOD16A2/A3) NASA Earth Observing System MODIS Land Algorithm. NASA: Washington, DC, USA.

Shamsudduha, M., Taylor, R.G., Jones, D., Longuevergne, L., Owor, M., Tindimugaya, C., 2017. Recent changes in terrestrial water storage in the Upper Nile Basin: an evaluation of commonly used gridded GRACE products. Hydrol. Earth Syst. Sci. 21 (9), 4533–4549. https://doi.org/10.5194/hess-21-4533-2017.

Swenson, S., Wahr, J., 2009. Monitoring the water balance of Lake Victoria, East Africa, from space. J. Hydrol. 370 (1–4), 163–176. https://doi.org/10.1016/j.jhydrol.2009.03.008.

Thiery, W., Davin, E.L., Seneviratne, S.I., Bedka, K., Lhermitte, S., Van Lipzig, N.P., 2016. Hazardous thunderstorm intensification over Lake Victoria. Nat. Commun. 7 (1), 1–7. https://doi.org/10.1038/ncomms12786.

Thiery, W., Gudmundsson, L., Bedka, K., Semazzi, F.H., Lhermitte, S., Willems, P., Lipzing, N.P.M., Seneviratne, S.I., 2017. Early warnings of hazardous thunderstorms over Lake Victoria. Environ. Res. Lett. 12 (7), 074012 https://doi.org/10.1088/1748-9326/aa7521.

USDA (United States Department of Agriculture), 2022. Global Reservoirs and Lake Monitor (G-REALM). U.S. Department of Agriculture Foreign Agriculture Service. Available: ⟨https://ipad.fas.usda.gov/cropexplorer/global_reservoir/⟩.

USGS (United States Geological Survey), 2018. Digital Elevation-Shuttle Radar Topography Mission (SRTM). USGS Earth Resources Observation and Science (EROS) Archive [Dataset].

Vanderkelen, I., van Lipzig, N.P., Thiery, W., 2018. Modelling the water balance of Lake Victoria (East Africa)–part 1: observational analysis. Hydrol. Earth Syst. Sci. 22 (10), 5509–5525. https://doi.org/10.5194/hess-22-5509-2018.

Wainwright, C.M., Finney, D.L., Kilavi, M., Black, E., Marsham, J.H., 2021. Extreme rainfall in East Africa, October 2019–January 2020 and context under future climate change. Weather 76 (1), 26–31. https://doi.org/10.1002/wea.3824.

WMO (World Meteorological Organization), 2017. WMO Guidelines on the Calculation of Climate Normal. World Meteorological Organization. Available: ⟨https://library.wmo.int/doc_num.php?explnum_id=4166⟩.

World Bank Data Catalog, 2019. Major River Basins of The World. The World Bank International Bank for Reconstruction and Development (IBRD) and International Audit Vice Presidency (IAD) [Dataset].

WWF (World Wildlife Fund), 2019. HydroSHEDS, DEG, WWF Water Risk Filter. Hydrology in Africa. Esri Map Service [Dataset].